

Degrees of separation: where the academy’s hiring order and graduates’ pay agree

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Abstract

Universities sell two products under one name: doctoral graduates, whom the academy hires as faculty, and bachelor’s graduates, whom employers hire. Faculty hiring reveals an ordering of departments, and graduates’ pay an ordering of programs. We ask where these two orderings of the same institutions agree, field by field. Using public data only, we measure within-field coupling, the rank correlation across institutions between a department’s position in the US faculty-hiring hierarchy and its graduates’ median earnings, for 52 US fields (College Scorecard, with Census PSEO as an overlapping second source) and, with a UK hiring hierarchy built from ORCID-derived records, for 33 UK subjects (DfE LEO). A short model states when hiring reveals the academy’s valuation of a department and pay the labour market’s valuation of its graduates, and what coupling then measures. The analysis is descriptive. Where the two orderings agree, they agree at the institution’s name, mostly in what both share with the selectivity of its intake: controls for selectivity, Pell share, control and state lower mean US coupling from +0.43 to +0.14. About a fifth of pay coupling runs between employer sectors rather than within them. Below the name they agree little. Within institutions, one standard deviation of department prestige goes with about 0.8% higher four-year pay (up to about 3% corrected for error in the ranks), a remainder whose 95% interval reaches zero, individually clear only in computer science; field prestige does not detectably out-predict academia-wide prestige. In the UK most of the ordering survives a direct control for graduates’ prior attainment. Within fixed graduation cohorts coupling rises by about 0.03 per year in both countries; in the US about three fifths of the rise is shared with selectivity. Coupling is weak where pay follows national scales. The hiring ranks partly reflect doctoral production (an audit bounds that part from above), and the data do not say why agreement below the name is small.

Keywords: faculty hiring networks; academic prestige; graduate earnings; college selectivity; reputation; field of study; administrative data; science of science

1 Introduction

Each faculty hire records where the new professor’s doctorate came from. Aggregated over a decade, the net direction of these hires orders departments in a steep hierarchy in which a few institutions train most of the faculty in their field (Clauset et al., 2015; Wapman et al., 2022b): a net-placement position. The ordering comes from peers’ hiring decisions, not from admissions statistics, reputation surveys or composite rankings, and it exists field by field. What employers pay the same institutions’ bachelor’s graduates orders them a second time. A university thus sells two products under one name, doctoral graduates to the academy and bachelor’s graduates to the labour market, and each audience’s choices reveal an ordering. We ask where the two orderings agree: within a field, do graduates of departments higher in the hiring hierarchy earn more, and does the agreement belong to the department or to the institution around it?

The question matters for two literatures. For the science of science, the hiring hierarchy is a carefully measured status ordering whose positions shape productivity, the spread of ideas and who becomes faculty (Way et al., 2019; Morgan et al., 2018, 2022; Zhang et al., 2022); whether the market’s ordering of the same institutions agrees with it, outside academia, where nearly all bachelor’s graduates work, has not, to our knowledge, been measured. For the economics and sociology of higher education, it sharpens an old question. Earnings rise with institutional selectivity and brand, much of that gradient is attributed to who enrolls (Dale and Krueger, 2002, 2014; Mountjoy and Hickman, 2021), the gradient differs by field of study (Eide et al., 2016; Quadlin et al., 2021; Hastings et al., 2013; Bloem et al., 2024), and in the United Kingdom an institution’s academic standing adds little information about graduates’ returns beyond the selectivity of its intake (Britton et al., 2022). A department-level, peer-defined prestige axis asks how much of the status gradient in pay belongs to the department, and how the gradient changes over careers, the subject of employer-learning models (Farber and Gibbons, 1996; Altonji and Pierret, 2001).

We measure within-field coupling: the Spearman rank correlation, across the institutions that teach a field, between a department’s position in the US faculty-hiring hierarchy (Wapman et al., 2022b) and the median earnings of its bachelor’s graduates. We map it across 52 US fields on College Scorecard earnings (U.S. Department of Education, 2026), repeat it on Census Post-Secondary Employment Outcomes (PSEO) (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026), and, for the United Kingdom, build an academia-wide hiring hierarchy from ORCID-derived records of PhD-to-faculty moves (Li, 2026) and set it against Longitudinal Education Outcomes (LEO) earnings, which are also released by graduates’ prior attainment (Department for Education, 2026). All inputs are public, and every result is descriptive.

A short model (Section 3) reads faculty hiring as revealing a department’s net-placement position, which is the academy’s valuation only if hires flow down a single valuation and production-to-hiring volume does not set their direction, and program pay as the labour market’s valuation of a program’s graduates, selection included. Coupling then measures how far the two orderings agree, in a name, a department and a growth term. The model was written after most results were known; its only content not fitted to results already seen is nine tests, specified before the analysis but not publicly registered.

The findings come in three layers. *Relational* findings need only classical measurement error: the hiring order and the pay order of the same institutions agree to a degree that varies across fields, with a similar field ordering on the two US earnings sources; the agreement sits mostly at the name, in what field prestige shares with academia-wide prestige and with institutional selectivity; in the UK most of it survives a direct control for graduates’ prior attainment; it rises with years since graduation within fixed cohorts in both countries (in the US its career-time part can only be bounded); and it is weak in UK subjects paid on national scales. *Construct* findings add the model: where the academy’s and the market’s orderings agree, they agree at the institution’s name, mostly in what both share with the selectivity of its intake; below the name they agree little, clearly only in computer science. Controls for selectivity, Pell share, control and state lower mean US coupling from +0.43 to +0.14, and about a fifth of pay coupling runs between employer sectors, which counts as the market’s valuation only if employers ration access to sectors; within institutions one within-field standard deviation (SD) of department prestige goes with about 0.8% higher pay, an interval that reaches zero (up to about 3% corrected for error in the ranks); and a field’s own ordering does not detectably out-predict the academia-wide ordering. *Mechanism* claims are not made: why agreement below the name is small is left open.

Much of this is known in kind (Section 2). What is new is (i) the faculty-hiring hierarchy as the prestige axis, which separates a department’s standing in its field (field prestige, F) from its institution’s standing in the academy as a whole (academia-wide prestige, G), and with it a within-

institution test of whether higher-standing departments inside the same institution have better-paid graduates; (ii) a model that reads both indicators as revealed valuations of two audiences and names the assumptions that can fail; (iii) fixed-cohort estimates, in two countries, of how coupling changes with years since graduation, new in design and scope rather than in direction (Thomas and Zhang, 2005; MacLeod et al., 2017); and (iv) measurement: a reliability bracket and a production audit for the published ranks, a check against noise in the program medians, and a UK academia-wide hiring hierarchy. The US field map is a replication in kind: its ordering is close to the known ordering of fields by how steeply pay follows selectivity (Bloem et al., 2024).

For the academy’s departmental hierarchy the answer is mostly negative. The small remainder beyond institutional status cannot be attributed to department standing rather than program-level selection, and the computer-science estimate describes a sample without the University of California campuses and several flagship public departments (Section 5.2). Further methods and all robustness detail are in the Supplementary Information (SI).

2 Background and positioning

College quality and selection. Once application behaviour absorbs selection, the average earnings payoff to a more selective college is small, with larger returns for disadvantaged students (Dale and Krueger, 2002, 2014); other designs find sizeable effects in some settings (Hoekstra, 2009; Brewer et al., 1999; Witteveen and Attewell, 2017). Single quality proxies are noisy (Black and Smith, 2006), estimates depend on common support, the part of the distribution and the selection controls (Black and Smith, 2004; Andrews et al., 2016; Cunha and Miller, 2014), and the payoff to elite colleges concentrates in the upper tail (Zimmerman, 2019; Chetty et al., 2026). Most of this work measures standing by selectivity, rankings, brand or value added (Riehl et al., 2019; Mountjoy and Hickman, 2021). The closest academic measure is Hussain et al. (2009): in UK graduate surveys an institution’s research rating predicts wages but correlates 0.87 with entry tariff and turns negative, imprecisely, alongside other quality measures.

Field of study and status by field. Returns to field are large and heterogeneous (Kirkeboen et al., 2016; Altonji et al., 2012, 2016a; Webber, 2014; Kim et al., 2015; Andrews et al., 2024), and so is the status gradient within field (Eide et al., 2016; Quadlin et al., 2021; Hastings et al., 2013). Closest to us, Bloem et al. (2024) show on an earlier Scorecard release that median pay barely varies with admit rate in nursing, teacher education and social work but strongly in computer science, finance and economics; cross-classified models find selectivity matters more outside STEM (Muse and Muse, 2024). UK returns by institution and subject were estimated on surveys (Walker and Zhu, 2011, 2018) and now on administrative LEO records (Belfield et al., 2018; Britton et al., 2019, 2020). The closest design to ours is Britton et al. (2022): within subject, returns at age 30 track the prior attainment of a degree’s intake closely in law, business and economics and not at all in medicine or education, which the authors attribute to centrally regulated wages, and net of selectivity no subject’s league-table rating correlates positively and significantly with returns. Our field map and UK check, which uses a coarser, band-level attainment control, are replications in kind.

Faculty-hiring hierarchies. Centrality in PhD-exchange networks tracks departments’ reputational prestige (Burris, 2004); hiring networks are steeply hierarchical (Clauset et al., 2015; Wapman et al., 2022b); PhD origin is tightly linked to placement (Fowler et al., 2007; Headworth and Freese,

2016); placement records have been used to rank departments (Amir and Knauff, 2008); and networks divide into clusters with little hiring between them (Terviö, 2011). This literature measures standing within academia; to our knowledge it has not asked whether position in the hierarchy predicts the pay of the academy’s bachelor’s graduates, a question made less than obvious by the weak link between research and teaching quality (Hattie and Marsh, 1996).

Employer learning and career dynamics. In employer-learning models, the wage coefficient on a credential that employers see at hiring need not change with experience when a hard-to-observe correlate of productivity is left out (Farber and Gibbons, 1996); with that correlate in the model, the credential’s coefficient falls with experience while the correlate’s rises (Altonji and Pierret, 2001). Employers learn quickly on average (Lange, 2007), but the speed varies by occupation (Mansour, 2012), and learning about a productivity that keeps changing stays imperfect over whole careers (Kahn and Lange, 2014); for college graduates the loading on test scores is already high at entry and grows little (Arcidiacono et al., 2010). Among students who applied to and were admitted by similar colleges, the premium to a more selective one is small over the career (Dale and Krueger, 2014) or fades within a few years (Mountjoy and Hickman, 2021); prestige premia fade with experience in Chile (Bordón and Braga, 2020) and matter less for later promotions in Japan (Araki et al., 2016), while US graduates of higher-quality colleges have faster early earnings growth (Thomas and Zhang, 2005) and college reputation correlates with earnings growth in Colombia (MacLeod et al., 2017). Entry conditions and field-specific forces shape careers too (Oreopoulos et al., 2012; Roksa and Levey, 2010; Altonji et al., 2016b; Deming and Noray, 2020); separating career time from calendar time and cohort requires assumptions (Fosse and Winship, 2019a,b), and administrative earnings records lose graduates selectively (Foote and Stange, 2026).

3 Two audiences, one name: what the indicators and coupling measure

Why should faculty hiring measure the academy’s standing of a department, and program pay the labour market’s valuation of its graduates? This section states when each reading holds and what ρ_f then measures. We wrote the model after most results of Section 5 were known; it organises them, and they are not evidence for it. Its only content not fitted to results already seen is nine tests (T0–T8), specified before the analysis but not publicly registered, on data that overlap those of results already seen. Derivations, a mapping of 28 results to the propositions, and each test’s specification and result are in SI Section S11.

Two audiences, one name. A department in field f sells doctoral graduates to the field’s hiring committees (the academy) and bachelor’s graduates to employers (the market), both under its institution’s name. Following MacLeod and Urquiola (2015) and Tadelis (1999), a unit’s reputation with an audience is that audience’s expectation of the quality of what it buys from the unit. Beliefs are unobserved, but repeated costly choices reveal the orderings they imply: the audience’s valuation. Write the academy’s valuation of department (i, f) as $s_{if} = \alpha_f a_i + r_{if}$ and the mean productivity of its bachelor’s graduates, selection included, as $\theta_{if} = \mu_f + \beta_f x_i + y_{if}$, with institution components a_i and x_i , departmental deviations r_{if} and y_{if} , and β_f the reward of field f ’s labour market for the institution component. The two sides correlate within a level ($\text{Corr}(a, x) = \chi$, $\text{Corr}(r, y) = \kappa_f$) but, we assume, not across levels ($\text{Cov}(r, x) = 0$, which fails if departments are relatively strong at selective institutions).

Proposition 1a: what hiring reveals. Let A_{uv} count field- f faculty at v with a PhD from u . Under vertical sorting (H1), in every pair of departments that exchanges faculty more hires are expected to flow from the higher-valued to the lower-valued department, and the ordering that minimises upward hires then agrees with the academy’s valuation on every exchanging pair. But H1 holds even without valuation: under random matching, flows scale with production and hiring, and departments are ordered by their ratio of PhDs produced to faculty hired. Volume neutrality (H0), that this ratio does not set the direction of flows, is what makes the direction informative. The published F is SpringRank (De Bacco et al., 2018), which has no degree correction and ranks net exporters higher at equal valuation. F is therefore a department’s net-placement position. It reads as the academy’s valuation, academic prestige or status in a deference network (Burris, 2004; Podolny, 1993), only under H0 and H1, and it may partly reinforce itself (Long et al., 1979). Test T0 audits H0 and finds that F partly reflects production volume; test T1 asks whether one index orders departments both as producers and as employers of PhDs and finds that it does not, which bears on H1 (Section 4.1). We therefore do not claim the valuation reading: F is described as a net-placement position, not as one belief about the quality of a department’s PhDs.

Proposition 1b: what pay reveals. Suppose that employers compete for graduates and pay their posterior expectation of productivity, see institution and major at hiring and set pay freely rather than by a schedule, and that program medians cover the graduates who work for them. An employer’s weight on a unit’s own record, $\omega_u = n_u \tau_u^2 / (n_u \tau_u^2 + \sigma^2)$, grows with the number n_u of the unit’s graduates it has observed and attributed to it (τ_u^2 and σ^2 are the variances of unit means and of individual performance). A name pools the graduates of all of an institution’s programs, so ω_{name} is near one wherever employers see an institution’s graduates often; a department’s record counts only where employers attribute performance to departments. With the name’s mean known, the program’s log median t years after graduation is

$$m_{if}(t) = \mu_f(t) + \beta_f x_i + \Lambda_{ft} y_{if} + g_i(t), \quad \Lambda_{ft} = \omega_f + (1 - \omega_f) \lambda_t, \quad (1)$$

where $\lambda_t = b + (1 - b) t / (t + k)$ is the weight on a graduate’s own revealed performance (b at hiring, k the speed of learning) and $g_i(t)$ is growth that differs across institutions. Program pay is then the labour market’s valuation of a program’s graduates. At the name, employers’ belief and the realised mean productivity of the institution’s graduates coincide, selection included. We reserve *employer reputation* for what the design of MacLeod et al. (2017) identifies, a school-mean premium with graduates’ own attainment held fixed, which only the UK attainment bands identify, and coarsely (Section 5.3). Three further limits apply: raw pay contains local price levels and access to high-wage labour markets; placement into better-paid sectors counts as the market’s valuation only if employers, not graduates’ preferences, allocate it; and where pay follows a schedule the wage stops revealing valuation (Proposition 5).

Proposition 2: what coupling measures. With $F = s + e_F$, classical errors and no covariance across levels,

$$\text{Cov}\{F_{if}, m_{if}(t)\} = \underbrace{\alpha_f \beta_f \chi \sigma_a \sigma_x}_{\text{name}} + \underbrace{\Lambda_{ft} \kappa_f \sigma_r \sigma_y}_{\text{department}} + \underbrace{\alpha_f \text{Cov}\{a_i, g_i(t)\}}_{\text{growth}}. \quad (2)$$

G pools the hires of all fields, so it carries only the name and growth terms. For Spearman correlations on the same institutions, $c_F = r_{FG} c_G + \sqrt{1 - r_{FG}^2} \text{sr}_F$ is an identity, where c_X is the coupling of X with pay, r_{FG} the correlation of F with G and sr_F the semi-partial correlation

of pay with the part of F orthogonal to G . F out-predicts G only if $\text{sr}_F > c_G \tan(\vartheta/2)$, where $\cos \vartheta = r_{FG}$; at the across-field means ($r_{FG} = 0.79$, $c_G = 0.45$) the threshold is 0.16 and sr_F is 0.08, a gap the model accommodates but does not imply. Three readings follow. First, coupling measures agreement between two audiences' orderings, not agreement on what they value: an academy that values research and a market that values graduates' ability agree whenever the two covary across institutions ($\chi > 0$), a primitive, not a result. Second, if one institution factor dominates, each field's coupling is proportional to its β_f , and the field map shows how steeply each field's pay follows the institution factor that status indices share, whatever status index is used; the construct reading of F then matters only below the name. Test T5, which replaces F by faculty salaries, spending or endowment, is inconclusive. Third, average SAT and the admission rate are noisy measures of intake ability, part of x_i ; for the question whether the academy's standing carries information that the market prices beyond intake, they are confounds.

Proposition 3: below the name. At entry, if employers see nothing of a graduate beyond the credential ($b = 0$), the department term is $\omega_f \kappa_f \sigma_r \sigma_y$: it needs both that department standing carry information about graduates' productivity ($\kappa_f \neq 0$) and that employers attribute performance to departments ($\omega_f > 0$). After entry it needs only $\kappa_f \neq 0$, because each graduate's own record prices the department deviation ($\Lambda_{ft} \geq \lambda_t$). A small department term fits two readings: department standing carries little information about graduates' market productivity (content), or employers cannot see departments and learn slowly about individuals (information). If the learning speeds estimated in other populations (Lange, 2007; Arcidiacono et al., 2010) applied here (k near 3 gives $\lambda_4 \approx 0.57$), a small four-year remainder would imply little departmental content; nothing in these data tests this, no test here separates the two readings, and we make no claim about the mechanism.

Proposition 4: experience. (a) If employers' posterior mean for a name equals the name's mean ($\omega_i = 1$), learning about individuals cannot move program-mean pay, the result of Farber and Gibbons (1996) applied to program means. More generally, the loading of pay on a status measure net of selectivity falls with experience, the signature of Altonji and Pierret (2001), only if beliefs about names are imprecise ($\omega_i < 1$) and employers' entry prior has a status component that the measure tracks beyond selectivity. A name's record pools many graduates (SI Section S11), so under rational expectations ω_i is near one wherever employers see an institution's graduates often, and there the absence of a decline is close to guaranteed; it says nothing about learning about individuals. (b) A rise in the institution-level covariance of pay with status requires status-correlated growth, such as access to firms with internal ladders (Gibbons and Waldman, 1999; Rivera, 2015; Weinstein, 2018); learning about rarely seen names from a prior that under-weights status; migration toward high-wage labour markets; or a change in who is observed (on standardised scales, also shrinking pay variance unrelated to status). The shape of the rise does not separate learning from growth; where it occurs could (test T3, Section 5.4).

Proposition 5: pay schedules. If a share $\bar{\pi}_f$ of a field's graduates is paid on a schedule whose level where they work is $\bar{\tau}$, then to first order $\text{Cov}(F, m) = (1 - \bar{\pi}_f) \text{Cov}_{\text{market}}(F, m) + \bar{\pi}_f \text{Cov}(F, \bar{\tau})$, plus a term for sorting out of scheduled jobs. Under a national schedule the second term is zero at entry and can grow over careers only if promotion depends on status; exit that depends on status moves the sorting term. Weak coupling there is not evidence that the market's ordering disagrees with the academy's; the model's content is a prediction about quantities, such as where graduates are hired (test T6, Section 5.5).

What the model licenses. Claims come in three layers. *Relational* claims need only classical measurement error: the hiring order and the pay order of the same institutions agree to a degree that varies across fields, with a similar field ordering on the two US earnings sources; the agreement sits mostly at the name, in what field prestige shares with academia-wide prestige and with institutional selectivity; in the UK most of it survives a direct control for graduates’ prior attainment; it rises with years since graduation within fixed cohorts in both countries (in the US its career-time part can only be bounded); and it is weak in UK subjects paid on national scales. *Construct* claims add Proposition 1: where the academy’s and the market’s orderings agree, they agree at the institution’s name, mostly in what both share with the selectivity of its intake; below the name they agree little, clearly only in computer science (about 0.8% of pay per SD of department prestige, up to about 3% under the lower-bound reliability of F). These claims read F as a net-placement position that is partly production volume, and pay as a valuation that includes selection. Pay is not netted of destination location: five-year coupling on the PSEO panel is unchanged when the wage level of graduates’ destination census divisions is netted out (ratio 1.01 [0.88, 1.15]; T7), but the Scorecard estimates (the control ladder, the partial couplings and the within-institution slope) were not re-estimated net of destination location, location within divisions is not netted, and a fifth to a third of the career rise moves with destination wages. About a fifth of pay coupling runs between employer sectors (0.21 [0.08, 0.34]; T8), which counts as the market’s valuation only if employers ration access to sectors. The claim below the name also assumes no covariance across levels, which the within-institution design relaxes only as far as SAT, admission rate, Pell share and G span the institution component x_i . *Mechanism* claims are not made: nothing here says what attending an institution or department does to pay, that wages equal productivity, that employers read academic rankings (the name term can be a pure common cause), or why agreement below the name is small. In the mapping of SI Section S11 no result follows for every admissible parameter value; half (14 of 28) are accommodated by parameter values inferred from the same results, and three are in tension with the model: one year after graduation field prestige carries the association and G does not; US special education and UK allied health couple strongly despite pay schedules; and across PSEO Flows cells the prestige and G gradients of pay are not robustly flatter where more graduates enter education, health care and public administration.

4 Data and measurement

Table 1 lists each source with its unit, the sample used and its known limitations; versions, URLs and access dates are recorded in the repository (SI Section S1).

4.1 Prestige: the faculty-hiring hierarchy

We use the published prestige ranks of Wapman et al. (2022b), estimated with SpringRank from a census of faculty at 368 PhD-granting US institutions, 2011–2020 (Wapman et al., 2022a). For institution i and field f , field (department) prestige F_{if} is minus the published field rank, and academia-wide prestige G_i is minus the rank in the network that pools hires across all fields. Both are fixed across earnings horizons and describe hires of 2011–2020. G tracks selectivity: across the 242 universities of the Opportunity Insights sample its Spearman correlation with average SAT is +0.68, and the UK analogue correlates +0.71 with intake selectivity across 133 providers. Within the field-by-cohort cells of the PSEO panel F and G correlate 0.83 on average.

We keep the labels field and academia-wide prestige, but two tests of Section 3 qualify what F measures. F is partly production volume (T0): replacing it by a degree-corrected Bradley–Terry rank keeps $R_{DC} = 0.84$ [0.74, 0.94] of mean coupling, and residualising it on log research

doctorates and the net export ratio of hires keeps $R_{\perp} = 0.60$ [0.47, 0.73] (56 fields). The reading fixed in advance is “partly production”. Because the residualisation also removes net export that valuation itself produces, $1 - R_{\perp}$ bounds the production share from above. F is also not one index for both roles (T1). With ranks fitted on one half of the hires and roles scored on the other, departments’ orderings as placers and as hirers of PhDs agree less than SpringRank’s own generative model produces under one index: the ratio of cross-role to within-role agreement is 0.658 against a benchmark of 0.828, a difference of -0.169 [-0.229 , -0.115] (33 fields), under a calibration of the benchmark chosen after the literal one had degenerated. The simpler role-consistency ratio also fails its threshold set in advance (0.525 [0.461, 0.583] against 0.80). We therefore read F as a net-placement position that is partly production, not as one belief about the quality of a department’s PhDs (SI Section S2, SI Section S11).

The public release omits institution pairs linked by a single hire, so a SpringRank rebuilt from public edges reproduces the published field ranks only partly (median Spearman 0.73; academia-wide 0.908). We therefore use the published ranks and bracket their sampling reliability: the split-half reliability of F on public edges averages 0.81 across 57 fields (stepped up with the Spearman–Brown formula; [Spearman, 1910](#); [Brown, 1910](#)), a lower bound, and 0.96 when extrapolated to the number of hires behind the published ranks; that of G is 0.99 (SI Section S2). Where reliability matters, results are reported at both ends.

4.2 Earnings and placement

US: College Scorecard. The canonical outcome is the median earnings of Title IV–aided bachelor’s completers four years after completion, from the Field-of-Study file released in June 2026 ([U.S. Department of Education, 2026](#)), which excludes completers enrolled or not working in the measurement year, aggregated over each field’s four-digit CIP codes through a curated map of 66 fields. Matching to the published ranks by normalised institution name gives 3,444 institution-by-field cells in the 52 fields with at least 15 matched institutions (median 48, range 15–173), the canonical sample, and 3,500 cells in the 57 fields with at least 8, used for heterogeneity, reliability and cross-source statistics. Scorecard covers 66–74% of the ranked institutions in each academia-wide prestige quartile (SI check, SI Section S1), but the name join loses institutions whose Scorecard names carry campus qualifiers (Georgia Tech, Purdue and Ohio State among them). Institutional covariates (average SAT, admission rate, Pell share, control, institution-wide earnings and the state earnings level) come from the “most recent” institution file, which describes recent entering classes; its SAT average is missing for test-blind institutions (the University of California campuses and Caltech among them), and for three key statistics we re-date the controls to the graduates’ entry years with the historical files. No public US file measures the entry attainment of a program’s own students. One- and five-year medians refer to other cohorts; they serve as replications and enter averages over the three horizons.

US: PSEO. PSEO reports unemployment-insurance wage earnings of graduates with earnings above a full-time minimum-wage threshold, in constant 2023 dollars, one, five and ten years after graduation (y1, y5, y10) by three-year graduation cohort. Release V4.13.0 (2025Q4) is canonical; the August 2026 release V4.14.1 (2026Q2), on which the later analyses run, is a re-check. For career time we keep, within each field and cohort (graduates of 2001–03 to 2010–12), the institutions observed at all three horizons, at least 15 per cell: 106 institutions, 142 field-by-cohort cells and 39 fields on V4.13.0 (116, 145 and 40 on V4.14.1). A window-by-horizon surface adds later cohorts: 17 cells, from 2001–03 to 2019–21 at y1, to 2016–18 at y5 and to 2010–12 at y10. PSEO partners are mostly public: only 20 of the 93 top-quartile ranked institutions have a usable five-year cell on

V4.13.0 (26 on V4.14.1). PSEO Flows give each institution-by-CIP-2 cell’s employed graduates by employer sector (SI Section S7) and by census division of work (SI Section S6, SI Section S11).

UK: LEO and an academia-wide hiring hierarchy. The DfE provider-level LEO file reports median earnings by provider, CAH2 subject, graduating cohort and year after graduation (YAG 1, 3, 5), separately by prior-attainment band, sex and POLAR4 quintile (one characteristic at a time) (Department for Education, 2026). We use the cohorts 2013/14 to 2016/17 and the 33 subjects with at least 15 providers. For prestige we run SpringRank on 11,987 PhD-to-faculty moves between institutions in Great Britain in the ORCID-derived academic mobility edges (Li, 2026; ORCID, Inc., 2026), matched to providers through the ROR data dump (Research Organization Registry, 2026): 134 providers form the primary sample (SI Section S1). Russell Group members sit at a mean percentile of 0.86 against 0.43 for other providers, and the split-half reliability is 0.92. Validity is less certain: in the US, an academia-wide ranking rebuilt from the ORCID-derived edges agreed with the published ranks at Spearman 0.74 (SI Section S2). Field-tagged UK networks are too sparse to use, so the UK analysis has no department axis. Intake selectivity is the institution’s share of graduates who entered with at least 360 tariff points.

Other sources. Table 1 lists the other sources: Opportunity Insights tail outcomes, the 2023 American Community Survey (licensed and sector shares, and wage levels by state, census division and sector), research doctorates from the Survey of Earned Doctorates and IPEDS, state unemployment rates, and admission rules for computer-science majors coded from official university pages.

4.3 Coupling and inference

Coupling is $\rho_f = \text{Spearman}_i(F_{if}, Y_{if})$ across the institutions i that teach field f , with Y median earnings: a rank statistic that asks whether the prestige and pay orderings coincide, not how large the pay differences are. Partial couplings correlate the rank residuals of F and Y after regressing both on the ranks of the controls. The main text rests on three estimands: coupling and partial coupling averaged over fields; the within-institution slope β of log earnings on department prestige (Eq. 3), a pay difference; and the per-year change of coupling within fixed graduation cohorts. A field is flagged reliable when its earnings carry cross-institution signal beyond sampling noise, its interval excludes the band that noise alone would produce under perfect agreement, and it has at least 10 institutions; 20 of the 52 canonical fields pass, and the flag is a check, not a filter (SI Section S3).

Fields share institutions, so Scorecard means across fields use a joint institution bootstrap, conditional on the observed fields; PSEO statistics use the two-way (field, institution) cluster variance (Cameron et al., 2011), with the conservative crossed field-by-institution bootstrap beside it; within-institution slopes use an institution cluster bootstrap; UK intervals resample subjects and then providers; tests clustered by state use the wild cluster bootstrap (Cameron et al., 2008; Webb, 2023); field-level moderators are tested by field-label permutation and by resampling whole CIP-2 families. Brackets are 95% intervals. Where interval types disagree, an association counts as present only if its interval excludes zero under the most conservative variance reported for it. Four statements fail that rule and are reported as not established (SI Section S9 lists their intervals): the within-institution slope, with 2026 or with entry-year selectivity controls; the gap between the growth of the G and F loadings; and the UK per-cell partial of G net of selectivity on all-graduate medians. The canonical specification was fixed after the main analyses had been run and before drafting; alternatives form a specification curve (Simonsohn et al., 2020) (SI Section S4). The

Table 1: Data sources and samples. All sources are public; provenance is recorded in `data/raw/SOURCES.md` of the repository.

Source	Unit and release	Sample used here	Role	Known limitation
Faculty-hiring ranks (Wapman et al., 2022a)	institution $\times$ field; published SpringRank ranks, 2011–2020	66 fields; 52 with ≥ 15 institutions	F, G	single-hire pairs not public; reliability of F bracketed
College Scorecard, Field of Study (U.S. Department of Education, 2026)	institution $\times$ CIP-4; bachelor’s median earnings at 4 years (1, 5 as replications); June 2026	3,444 cells, 52 fields	Y (canonical)	Title IV completers; medians; no program-level entry attainment
College Scorecard, institution files (U.S. Department of Education, 2026)	“most recent” (2026) and historical files (2001–02 to 2016–17)	46–47 fields with SAT data	selectivity, Pell, control, state	institution-level; entry year inferred
PSEO earnings (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026)	institution $\times$ CIP-4 $\times$ 3-year cohort; p25, p50, p75 at y1, y5, y10; V4.13.0, V4.14.1	fixed cohorts 2001–2012; 17-cell surface, 24 fields	second source; career time	public-skewed; 20 (26) of 93 top-quartile institutions
PSEO Flows (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026)	institution $\times$ CIP-2 $\times$ employer sector or census division of work; V4.13.0, V4.14.1	sectors: 1,312 cells, 108 institutions; divisions: fixed-cohort panel (T7), 95 and 116 institutions (T6)	placement; destination location	noisy counts; industry, not occupation; CIP-2 only
DfE LEO provider file (Department for Education, 2026)	provider $\times$ CAH2 $\times$ cohort $\times$ YAG; by prior attainment, sex, POLAR4	33 subjects, 134 providers	UK earnings	marginal breakdowns only; coarse bands
ORCID-derived mobility edges (Li, 2026); ROR (Research Organization Registry, 2026)	PhD-to-faculty moves within Great Britain	11,987 moves, 551 institutions	UK G	self-reported careers; academia-wide only
Opportunity Insights (Chetty et al., 2020)	college; outcomes by parent income; Barron’s 2009 index	242 universities	tail outcomes	pools a college’s fields
ACS PUMS 2023 (U.S. Census Bureau, 2026)	person	field, field $\times$ state, census-division and sector aggregates	shares; local demand; destination and sector wage levels	state of residence; nominal wages
Survey of Earned Doctorates (National Center for Science and Engineering Statistics, 2026); IPEDS (National Center for Education Statistics, 2026)	baccalaureate institution	PSEO panel institutions	doctorate rate	institution-wide
BLS LAUS (U.S. Bureau of Labor Statistics, 2026); admissions pages	state $\times$ year; CS program	18 states; 88 of 128 programs	entry conditions; entry route	residence state; 2026 policy
Occupational prestige (Hughes et al., 2024); NY Fed data by major (Federal Reserve Bank of New York, 2026)	field	32–52 fields	placement check	field level only

analyses added later (scripts 65–69 and 72–75) wrote their tests and reading rules into the script before estimation, a self-reported timing; the nine tests of Section 3 (script 75) were specified before the analysis, after most other results had been seen, and were not publicly registered. SI Section S9 lists every hypothesis examined in scripts 52–75 with its outcome. Every number comes from a seeded script or, where marked “SI check”, from code run on the repository’s tables.

5 Results

Each subsection states a relational finding, which needs only classical measurement error, and the reading that Section 3 licenses. Table 2 collects the controls for institution-level status and Table 3 the career-time estimates.

5.1 The two orderings agree, to a degree that varies across fields

In most fields the academy’s hiring order and the market’s pay order of the same institutions agree. Across the 52 canonical fields mean coupling is $+0.419$ [$+0.355$, $+0.482$], and 39 fields have coupling more than 1.96 bootstrap standard errors above zero, none below (Fig. 1). Coupling varies across fields by more than institution sampling produces: in a three-level meta-analytic model of Fisher- z coupling on 57 fields (Konstantopoulos, 2011; Cheung, 2014), Cochran’s $Q = 183.2$ on 56 degrees of freedom and $I^2 = 0.69$ (Higgins and Thompson, 2002). Propagating noise in the program medians, in a secondary analysis not yet independently verified, lowers I^2 to 0.51 and keeps the field ordering (SI Section S3). Part of the remaining variation is a measurement confound: coupling rises with the number of institutions that teach a field (Spearman 0.43, 57 fields), and a discipline (CIP-2) variance component, detected without adjustment, is borderline once field size is a covariate (permutation p from 0.028 to 0.046; SI Section S3). The grouping of fields into disciplines is therefore a descriptive map.

The field ordering is similar on a second earnings source: coupling from PSEO five-year earnings correlates with Scorecard coupling across fields at $+0.68$ (two-way interval [$+0.54$, $+0.82$]; 57 fields, V4.13.0) and $+0.74$ [$+0.61$, $+0.87$] on V4.14.1. This is robustness to the earnings measure, not an independent replication: both use the same F , and 61% of the PSEO rows used on V4.14.1 also enter the Scorecard couplings.

Two checks bear on what the pay side measures. Net of the wage level of the census divisions where graduates work, five-year coupling on the PSEO panel is unchanged (ratio to raw coupling 1.01 [0.88, 1.15], 39 fields; T7); location within divisions is not netted. Pay also tracks where graduates go: within CIP-2 families in PSEO Flows, family prestige orders the share of employed graduates in market sectors about as closely as it orders pay ($+0.40$ [$+0.32$, $+0.47$] against $+0.41$ [$+0.31$, $+0.49$]; 1,312 cells; SI Section S10), and about a fifth of the pay coupling runs between employer sectors (0.21 [0.08, 0.34]; T8). That part counts as the market’s valuation only if employers, not graduates’ own choices, allocate access to sectors, which these data cannot show.

5.2 Where the orderings agree: at the institution’s name

Where the academy’s and the market’s orderings agree, they agree at the institution’s name, mostly in what both share with the selectivity of its intake; below the name they agree little, clearly only in computer science. Within institutions one SD of department prestige goes with about 0.8% of pay, and up to about 3% under the lower-bound reliability of F , so the department-specific part is diluted by the name, not shown to be absent. The pay in these statements is Scorecard pay, not netted of destination location, and about a fifth of pay coupling runs between employer sectors (T8; Section 5.1), which counts as the market’s valuation only if employers ration access to sectors. The estimand is the information in F about pay net of institution-level status proxies, not a return. Average SAT is a candidate quality treatment as well as a selection proxy (Dale and Krueger, 2002), so the drop measures what public data cannot separate from institutional status, not what selectivity causes.

The control ladder. On a common sample of 44 fields, mean coupling falls from +0.43 raw to +0.31 with Pell share, control and the state earnings level, to +0.24 with the admission rate and to +0.14 with average SAT (Fig. 2a; Table 2). On the 46 fields estimable with the broad set (SAT, admission rate, Pell share, control and state earnings level) the mean partial coupling is +0.141 [+0.075, +0.208], and 11 fields stay more than 1.96 standard errors above zero. Dating the controls to the falls in which the earnings cohort entered (2014–15) leaves the broad partial where it was (difference +0.005 [−0.031, +0.041] on the same 2,787 programs, two-way; SI Section S4). Adding G to the broad set leaves +0.059 [+0.003, +0.114] (44 fields), the part of the department ordering that neither selectivity nor the academia-wide ordering tracks; across ten specifications that net out selectivity and G , its interval excludes zero in four.

Within institutions. Institution fixed effects absorb what is common to an institution’s programs, including the institution-wide parts of selectivity, G and location. The slope of log four-year earnings on department prestige falls from +0.084 (SE 0.006) to +0.013 per within-field SD when they are added (3,416 programs, 198 institutions; Fig. 2b). With field-specific slopes on SAT, admission rate, Pell share and G (Eq. 3) it is +0.0084 [+0.0000, +0.0164] (2,889 programs, 168 institutions, 47 fields; institution cluster bootstrap; [−0.0012, +0.0180] under the two-way variance): one SD of department prestige goes with 0.84% higher four-year pay, with an upper 95% limit of 1.6–1.8% and a lower limit at or below zero. With one- and five-year medians, which describe other cohorts, the slope is about zero, and averaged over the three horizons its interval includes zero unless programs are weighted by graduates (Table 2). The point estimate barely moves with selectivity dated to entry years (+0.0081 against +0.0090 on the same 2,801 programs), cubic field-specific slopes in G or a field-by-state control for local demand, a weak control that describes the institution’s state (SI Section S4). A field-specific control for the program’s own Pell share does not lower it either (Table 2); with that control the slope is smaller for aided graduates without a Pell grant than for Pell recipients, a hint of composition within programs that the partial-correlation design does not repeat (SI Section S4).

Department prestige against its competitors. In per-field rank regressions on 47 fields, the mean Shapley share of R^2 (Budescu, 1993; Grömping, 2007; Shorrocks, 2013) is 0.076 for F , against 0.173 for selectivity, 0.086 for G and 0.201 for institution-wide earnings, which contain the field’s own graduates (Fig. 2c). Before any control, G couples with field earnings slightly more strongly than F (mean $c_F - c_G = -0.049$ [−0.085, −0.011], 57 fields), but F is measured with more error. By the identity of Proposition 2, F beats G only where the part of F orthogonal to G correlates with pay by more than $c_G \tan(\vartheta/2)$, which happens in 15 of 57 fields. Corrected for reliability (Spearman, 1904), the difference is −0.005 [−0.043, +0.044] at the lower-bound and −0.042 [−0.077, −0.001] at the extrapolated reliability, and net of selectivity it is +0.00 [−0.05, +0.05] (Fig. 2d). Field prestige thus does not detectably out-predict academia-wide prestige under any correction, although the upper 95% limits (+0.044 under the lower-bound reliability, +0.05 net of selectivity) leave room for a small advantage; whether it predicts pay worse depends on its reliability.

Where the remainder sits. Only computer science, one of three fields named in advance as high-coupling, has an individually clear remainder: +0.050 [+0.012, +0.096] per SD pooled over horizons (128 programs; [+0.007, +0.114] correcting for the three fields). Whether per-field slopes differ at all depends on whether each field’s earnings gradient in G may be curved, and the concentration in computer science and engineering was read off the results (SI Section S4). The

computer-science sample lacks 21 of the 50 highest-ranked departments: the six University of California campuses and Caltech, which have no fall-2024 SAT average; 11 lost in the name join, among them Georgia Tech, Purdue, Ohio State and the University of Washington; and 3 without a one- or five-year median. Entry-year SAT averages and a join by institution identifier would restore most of them, but neither re-estimate was run, so the computer-science result describes this sample only. A comparison of programs that restrict entry to the major with programs that students can declare is too weak to separate selection into the major from department standing (SI Section S4).

Two readings. The residual prestige measure is noisy, and partialling G removes some genuinely field-specific variation, so the remainder is biased toward zero: disattenuated with the reliability bracket, the within-institution slope is $+0.0098$ to $+0.0297$. A coefficient-stability bound (Oster, 2019) puts the selection on unobservables needed to erase it at 0.20–0.67 times the selection on the observed controls ($R_{\max} = 1$), below the usual benchmark of 1 (SI Section S4). Under one reading, the academy’s field-specific hierarchy carries labour-market information of its own; under the other, coupling is the field’s gradient in institutional status plus student quality that institution-level measures miss, for example selection into majors that admit on grades (Bleemer and Mehta, 2022). The bulk of the coupling fits the second; for the remainder, including the Pell/non-Pell contrast, the two predict the same pattern. The cross-field ordering does not separate them either: partial couplings still rank fields like raw couplings beyond shared estimation noise, but raw coupling and how steeply each field’s earnings follow SAT have a true-score correlation of $+0.94$ (23 fields; SI Section S4), which one dominant institution factor would produce (Proposition 2). Nor does the upper tail: net of selectivity and parental income, G ’s association with the share of former students in the top 1% does not detectably exceed its association with median earnings (242 universities; SI Section S8).

5.3 UK: most coupling survives a direct prior-attainment control

The UK data add what the US data lack, earnings by graduates’ own prior attainment, but they have no department axis, so they bear on institution-level status against student attainment, not on the department question.

UK coupling of G with LEO earnings five years after graduation averages $+0.486$ [$+0.398$, $+0.559$] across 33 subjects, similar to the US coupling of G with Scorecard earnings ($+0.450$, 57 fields). Most of it survives a control for attainment (Table 2, panel C; Fig. 3). Among same-band graduates across providers coupling is $+0.227$ against $+0.342$ for all graduates of the same providers (28 subjects), so 66% [51, 79] survives, and 87% [74, 100] once the lower reliability of band medians is corrected for (27 subjects). Standardising each provider’s earnings for its band mix, with band gradients measured within providers, leaves 85% [80, 90] (32 subjects). These shares describe orderings; in pay terms less survives, 73% [67, 78] of the log-earnings gradient in G . Coarse bands, sorting within a band, and within-provider gradients identified from students admitted despite a lower band all push the shares up (SI Section S5).

Among graduates of the same attainment band, what survives tracks intake selectivity at least as closely as hiring-network prestige if G is measured with little error: G adds $+0.033$ beyond selectivity (two-way [-0.047 , $+0.112$]), and selectivity keeps $+0.252$ [$+0.172$, $+0.310$] beyond G . On all-graduate medians G keeps more: net of selectivity $+0.131$ [$+0.078$, $+0.188$], 26% of the raw coupling, a fall comparable to the US one. The UK G may measure standing with error: at a validity of 0.74 the data allow a same-band partial up to $+0.26$ (SI Section S5). Subjects differ, and the differences survive the attainment control, but subjects whose pay couples strongly with G are those whose pay couples strongly with intake selectivity (Spearman $+0.92$ [$+0.71$, $+0.95$]); as

Table 2: Coupling under institution-level controls. Panels A and B: US, College Scorecard median earnings four years after completion unless stated, published field prestige F . Panel C: UK, LEO median earnings five years after graduation (cohorts 2013/14–2016/17), academia-wide UK hiring prestige G . k = fields (subjects). Intervals (95%): joint institution bootstrap, conditional on the observed fields (A); institution cluster bootstrap (B); two-stage bootstrap over subjects and providers (C); † : two-way (field, institution) or (subject, provider) cluster variance.

Specification	k / sample	Estimate [95% interval]
<i>A. US: mean within-field coupling of F with earnings</i>		
Raw, canonical sample	52	+0.419 [+0.355, +0.482]
Ladder (point estimates, common sample): raw $\rightarrow$ +	44	+0.43 $\rightarrow$ +0.31 $\rightarrow$ +0.24 $\rightarrow$
Pell share, control, state level $\rightarrow$ + admission rate $\rightarrow$ +		+0.14 $\rightarrow$ +0.12
SAT $\rightarrow$ + institution-wide earnings		
Broad set (SAT, admission rate, Pell share, control, state level)	46	+0.141 [+0.075, +0.208]
broad set dated to entry years (falls 2014–15) / 2026 file, same programs †	45; 2,787 programs	+0.134 [+0.065, +0.203] / +0.129 [+0.062, +0.197]
Broad set + G (field-specific remainder)	44	+0.059 [+0.003, +0.114]
Broad set + G + institution-wide earnings	44	+0.047 [−0.011, +0.106]
Broad set + G , mean of 4-, 1- and 5-year earnings	39	+0.052 [+0.003, +0.100]
<i>B. US: within-institution slope, change in log earnings per within-field SD of F</i>		
Field fixed effects only	3,416 programs	+0.084 (SE 0.006)
+ institution fixed effects	3,416 programs	+0.0129 [+0.0048, +0.0212]
+ field-specific slopes on SAT, admission rate, Pell share, G (main)	2,889 programs	+0.0084 [+0.0000, +0.0164]
same programs, field fixed effects only	2,889 programs	+0.0851 (point)
same programs, field-specific slopes on SAT and G only	2,889 programs	+0.0118 [+0.0037, +0.0205]
main + field-specific local demand (ACS field $\times$ state earnings)	2,889 programs	+0.0084 [+0.0002, +0.0165]
main, selectivity dated to entry years / 2026 file †	2,801 programs	+0.0081 [+0.0005, +0.0158] / +0.0090 [−0.0008, +0.0187]
main, 1-year earnings (4-year on the same programs +0.0123)	2,681 programs	−0.0024 [−0.0150, +0.0101]
main, 5-year earnings (4-year on the same programs +0.0111)	2,549 programs	+0.0023 [−0.0069, +0.0119]
main, mean of 4-, 1-, 5-year earnings: unweighted / weighted by graduates	2,422 programs	+0.0048 [−0.0024, +0.0120] / +0.0108 [+0.0035, +0.0181]
main + field-specific program Pell share, composition sample	1,294 programs	+0.0137 [+0.0027, +0.0238]
same, Pell-recipient / other aided graduates' median as outcome	1,294 programs	+0.0102 [−0.0024, +0.0222] / +0.0009 [−0.0123, +0.0128]
<i>C. UK: coupling of G with earnings and prior attainment</i>		
Raw	33	+0.486 [+0.398, +0.559]
Raw, composition sample	32	+0.497 [+0.406, +0.571]
standardised for band mix, within-provider gradients (85% [80, 90] survives)	32	+0.425 [+0.335, +0.497]
standardised for band mix, national band medians (floor; 62% [56, 69])	32	+0.310 [+0.234, +0.374]
log-earnings gradient per SD of G , raw $\rightarrow$ within-provider standardised (73% [67, 78]) †	32	+0.100 $\rightarrow$ +0.073 [+0.051, +0.095]
partial on institution intake selectivity (26% [17, 38]; two-way [+0.034, +0.229])	32	+0.131 [+0.078, +0.188]
Within prior-attainment band	28	+0.227 [+0.160, +0.279]
all graduates, same providers (66% [51, 79] survives; noise-corrected on 27 subjects 87% [74, 100], against 70% uncorrected there)	28	+0.342 [+0.255, +0.398]
within band: G net of intake selectivity †	28	+0.033 [−0.047, +0.112]
within band: intake selectivity net of G	28	+0.252 [+0.172, +0.310]

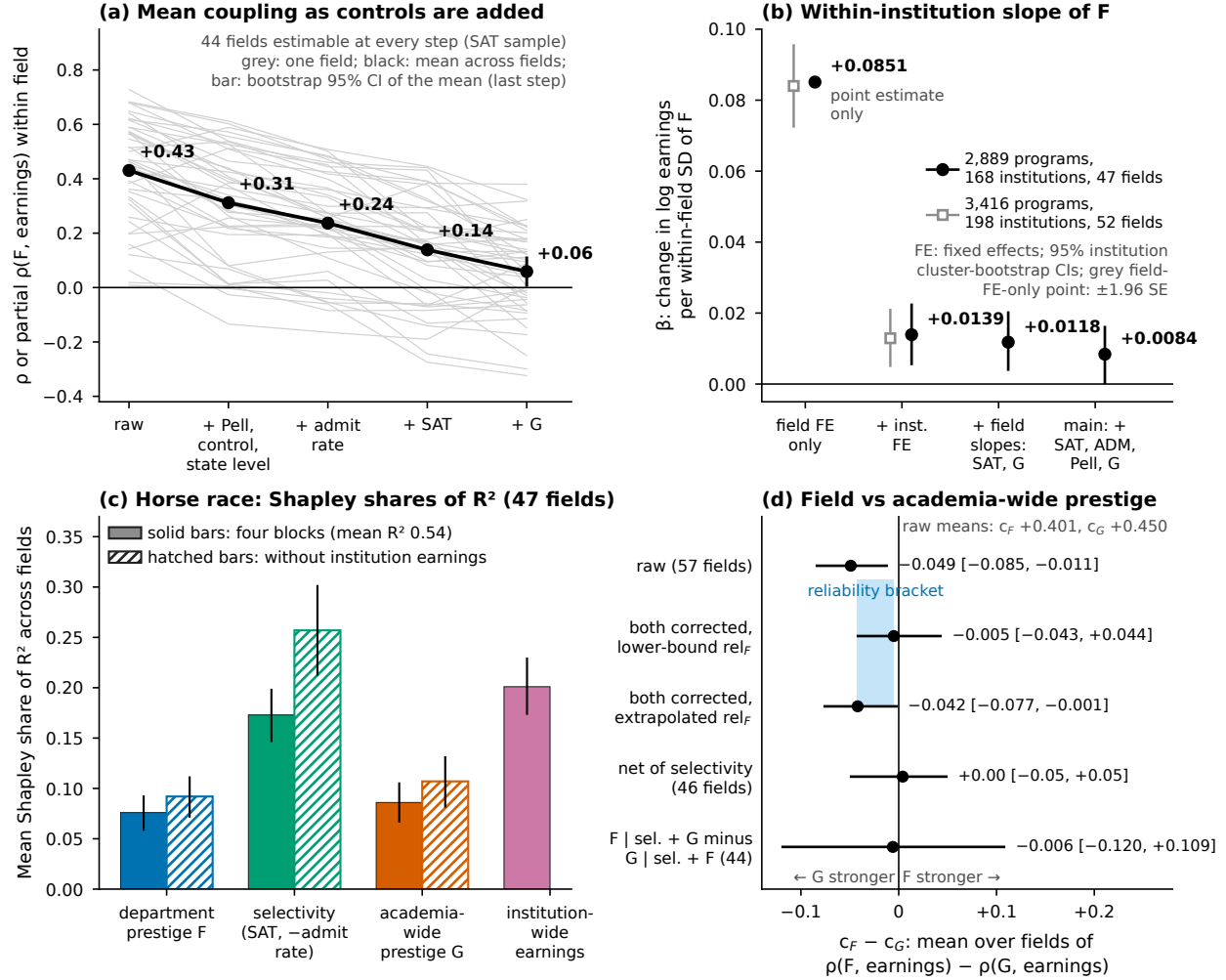


Figure 2: What remains of US coupling after institution-level status (Scorecard, four-year earnings). (a) Coupling of F as controls are added: fields (grey), mean (black; 44 fields). (b) Within-institution slope per within-field SD of F , from field fixed effects to the main specification (Eq. 3); black: 2,889 programs; grey: all 3,416. (c) Mean Shapley shares of R^2 in per-field rank regressions (47 fields), with and without (hatched) institution-wide earnings. (d) Mean $c_F - c_G$: uncorrected, corrected for error in F at either reliability, and net of selectivity. 95% institution-bootstrap intervals.

in the US, the subject ordering cannot be separated from subject-specific selectivity gradients (SI Section S5).

The within-band premium to intake selectivity net of G is the closest these data come to employer reputation in the sense of MacLeod et al. (2017). Within fixed cohorts and on average over bands it does not decay: its partial correlation rises by +0.027 [+0.016, +0.037] per year from YAG1 to YAG5 (T4; 27 subjects), or +0.0040 [+0.0027, +0.0054] log points per SD of selectivity per year, the opposite of what employer learning predicts if employers do not see attainment. Like the other UK within-cohort slopes, it mixes years since graduation with calendar time (Section 5.4). We therefore do not describe the premium as reputation that learning corrects. The rise sits in the middle bands, and the top band has cells only for medicine and dentistry, whose graduates are paid on national scales, so it says little about the band where own attainment is most uniform (SI

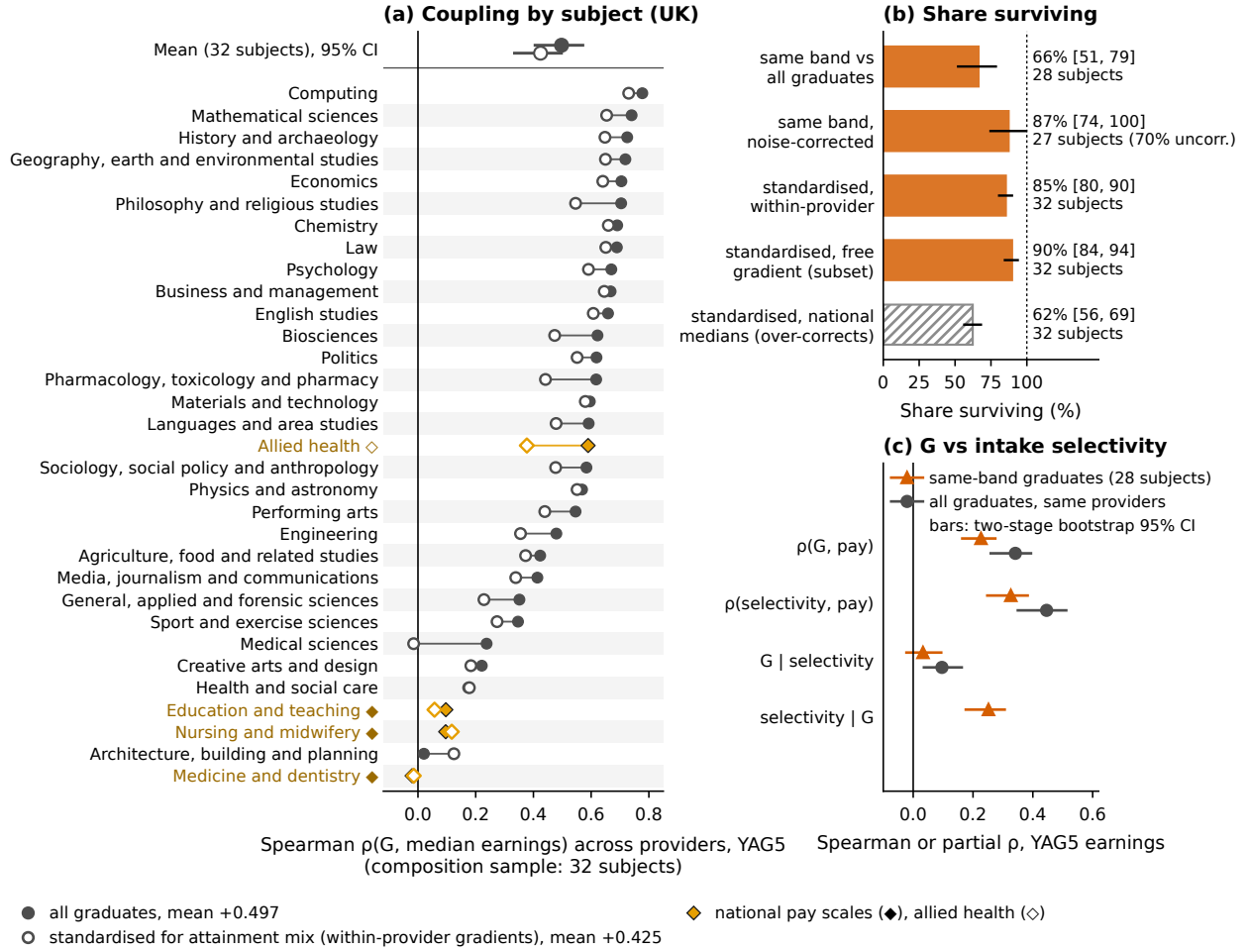


Figure 3: UK: coupling and prior attainment (LEO five-year earnings, cohorts 2013/14–2016/17). (a) Coupling of G by subject, all graduates (filled) and standardised for attainment mix (open), on the 32-subject composition sample, whose values differ from those on all providers (Fig. 5b). (b) Share of coupling surviving each attainment control. (c) Coupling of G and of intake selectivity with earnings, and each net of the other, for same-band (triangles) and all graduates (circles). Two-stage bootstrap 95% intervals; two-way intervals of $G \mid \text{selectivity}$: $[-0.047, +0.112]$ and $[-0.003, +0.195]$.

Section S11).

5.4 Agreement grows with years since graduation

Within fixed graduation cohorts, coupling rises with years since graduation in both countries. In the US it also rises across graduation cohorts, much of the within-cohort rise is shared with selectivity, and in a weakly identified decomposition it appears in the loading on academia-wide prestige G , not on field prestige F .

US within cohorts. On the balanced PSEO panel, mean coupling rises from +0.173 one year after graduation to +0.310 at five years and +0.451 at ten (39 fields, V4.13.0; Fig. 4a). The mean per-field slope is +0.031 per year (two-way interval $[+0.021, +0.041]$; +0.030 on V4.14.1, Table 3).

It is positive in each of the four cohorts and in both segments, so it does not rest on the first-year point, and a correction for sampling noise in the medians does not shrink it (exploratory; SI Section S6).

Cohorts and calendar time. A within-cohort change mixes career time with calendar time. On the 17-cell surface of graduation windows by horizon (24 fields whose institutions are present in every cell; V4.14.1), coupling rises both within windows ($\theta_h = +0.031$ [$+0.019, +0.043$] per year since graduation) and across windows at a fixed horizon ($\theta_c = +0.015$ [$+0.008, +0.023$]; Fig. 4b). Career, calendar and cohort trends are not separately identified (Section 8); if calendar and cohort trends are both non-negative, the career-time slope lies in $[\theta_h - \theta_c, \theta_h] = [+0.015, +0.031]$ per year (Fosse and Winship, 2019b), and the calendar-matched contrast gives nearly the same bound (Table 3). The bound also assumes that graduate study does not depress the early medians of higher-status institutions more than others'; under that confound both ends are biased upward. Prestige is measured once, from 2011–2020 hires, so any drift as cohorts approach that period enters b or g ; tests of entry conditions (cf. Oreopoulos et al., 2012) are not informative (SI Section S6).

Selection into the data, graduate study and location. Higher-prestige institutions have fewer graduates with PSEO earnings at y1; partialling coverage out removes about a fifth of the slope (Table 3; cf. Foote and Stange, 2026). Graduate study could raise coupling mechanically if graduates of higher-status institutions more often leave the earnings sample to study and return with higher pay. With coverage and the institution's research-doctorate rate per bachelor's graduate as controls, the rise is $+0.0103$ [$+0.0027, +0.0178$] per year, against $+0.0305$ raw on the same 144 cells; but the doctorate rate is institution-wide and tracks G (within-cell Spearman $+0.71$), so the removed share mixes timing with a status gradient. At the field level the mechanism is neither shown nor excluded (power 0.31). About one fifth to one third of the rise moves with the wage level of the census divisions where graduates work and with the out-of-state share (T7). The rise survives every available proxy; its size does not (SI Section S6).

Selectivity and status loadings. Net of SAT and admission rate dated to the cohorts' entry years, the within-cohort slope is $+0.011$ [$+0.003, +0.020$] per year, 0.39 of the raw $+0.029$ on the same 109 cells (V4.14.1): about three fifths of the rise is shared with selectivity. The loading of pay on G net of selectivity (the broad set, and F in the rank regression) does not decline within cohorts: its slope is $+0.016$ [$-0.011, +0.044$] per year in the rank regression, which excludes a decline steeper than 30% of the uncontrolled rise, and $+0.014$ [$+0.003, +0.024$] as a partial correlation (37 fields, V4.14.1). This test takes selectivity from the 2026 file, whose admission rates rank the PSEO institutions only loosely like those of their entry years (Spearman 0.45–0.47; SAT 0.88–0.90), a dating that favours finding no decline. In the UK, G 's loading net of selectivity is flat ($+0.008$ [$-0.006, +0.022$], two-way) while the selectivity loading rises ($+0.024$ [$+0.011, +0.037$]); their difference includes zero, and with a negative UK cross-cohort drift these are not career-time statements. By Proposition 4a, the absence of a decline is close to guaranteed under rational expectations about names that employers see often: it fits precisely known names, or an entry prior whose status component G does not track, and it does not bear on learning about individuals. The rising UK selectivity loading does not favour precisely known names; it fits learning about names, selectivity-correlated growth or a common rise of all standardised loadings (SI Section S6).

Academia-wide against field prestige. In a standardised rank regression of earnings on F and G within each field-by-cohort cell, b_F stays flat ($+0.189$ at y1, $+0.177$ at y10) while b_G grows from

-0.013 to $+0.328$ (Fig. 4d). Their slope difference $d_G - d_F$ is $+0.039$ [$+0.009$, $+0.069$] under the two-way variance (V4.13.0), but the conservative crossed bootstrap reaches zero, and the decomposition is weakly identified: within cells F and G correlate 0.83 , and G is measured more reliably. In a pooled form of the regression, and net of selectivity, G 's lead over F is not distinguishable from zero (SI Section S6). The rise of the G loading is not detectably faster or slower where employers are likely to see an institution's graduates less often: on V4.14.1 (40 fields), the G gradient rises about as fast at institutions with low as with high undergraduate enrolment given G ($+0.032$ and $+0.033$ per year; interaction $+0.0002$ [-0.0074 , $+0.0078$] per year, permutation minimum detectable effect 0.009 ; T3). What the growing G -tracked component is (residual selectivity, reputation, resources, location or the timing of graduate study) is not identified.

UK. On fixed LEO cohorts (33 subjects), coupling rises from $+0.379$ at YAG1 to $+0.484$ at YAG5, $+0.026$ per year [$+0.014$, $+0.037$] (Fig. 4c), in all four cohorts and both segments, and by $+0.028$ [$+0.016$, $+0.038$] after within-provider standardisation for prior attainment. The calendar-matched contrast ($+0.031$ [$+0.021$, $+0.042$]) exceeds the within-cohort change on the same providers ($+0.025$), so the sign assumption of the US bound holds only within sampling error, and offsetting calendar and cohort effects cannot be excluded (SI Section S5).

Reading. The direction agrees with descriptive evidence (Thomas and Zhang, 2005; MacLeod et al., 2017), and a descriptive gradient can rise while the causal selectivity premium is small or fades (Dale and Krueger, 2014; Mountjoy and Hickman, 2021). Given the three fifths shared with selectivity, the most parsimonious reading is residual selection, growing pay returns to pre-college differences that status and selectivity proxy; holding both fixed, institutions with more Pell students lose their early relative position (SI Section S6). The rise also fits the other sources of Proposition 4b, among them sorting into careers with steeper earnings profiles (Binder et al., 2016; Rivera, 2015), migration and the timing of graduate study; aggregate data cannot separate them.

5.5 Where agreement is weak: national pay scales and strictly licensed occupations

UK. Nursing and midwifery, medicine and dentistry, and education and teaching, three subjects whose graduates are largely paid on national scales, couple at $+0.120$ [-0.033 , $+0.258$] against $+0.523$ [$+0.439$, $+0.585$] for the other 30 subjects (difference -0.404 [-0.566 , -0.238]; exact subject-label permutation $p = 0.0009$; Fig. 5b), and the contrast survives standardisation for prior attainment (SI Section S5). Allied health, much of it also paid on national bands, is an exception at $+0.570$. The pattern and its reading as centrally regulated pay are Britton et al.'s (2022); we add nursing and a hiring-network axis.

US. Across US fields the prestige-pay association net of geography falls with the share of a field's graduates in strictly licensed occupations (health practitioners, school teachers, lawyers, architects and psychologists; ACS 2023): Spearman -0.53 to -0.48 over two institution samples and three geography codings (Fig. 5a). It is a contrast between discipline families (engineering, computer science, statistics and business at the low-share end), and it is fragile: under resampling of whole CIP-2 families its interval excludes zero in only 3 of 6 versions, it is not detected within families or with a broad licensure share, a government-or-non-profit employment share predicts it at least as well, and special education, mostly paid on district salary schedules, couples strongly. Pay set by the setting, such as a teacher salary schedule (Biasi, 2021) or a pay scale, is a reading consistent

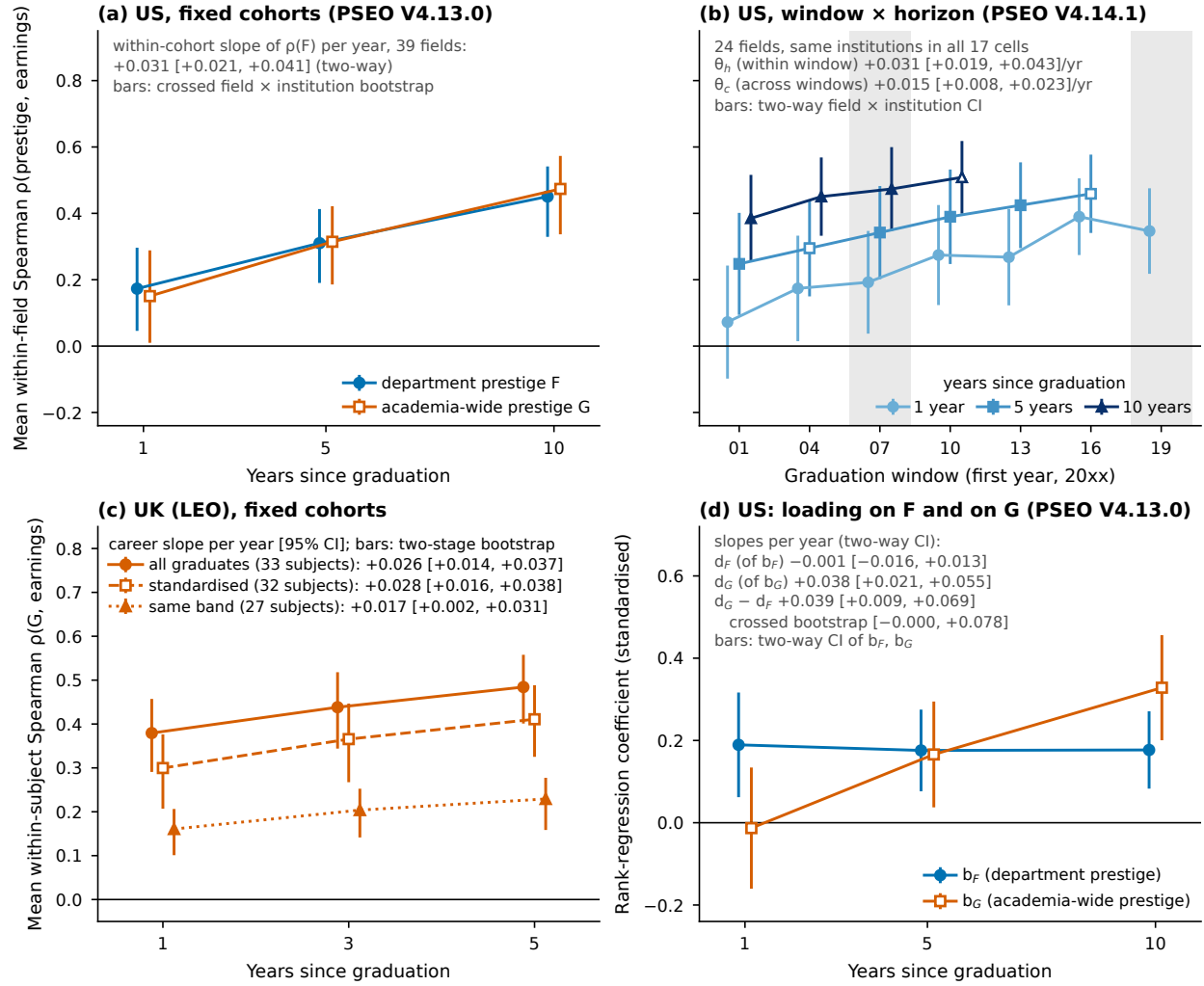


Figure 4: Coupling and years since graduation. (a) US: mean coupling of F and G with PSEO medians at one, five and ten years, fixed cohorts 2001–03 to 2010–12 (39 fields, V4.13.0). (b) US: coupling of F in 17 graduation-window $\times$ horizon cells (24 fields, V4.14.1); shading: earnings in 2008–10 or 2020–22. (c) UK: coupling of G , fixed LEO cohorts 2013/14–2016/17: all graduates, standardised for attainment mix, within bands. (d) US: rank-regression loadings on F and G , panel of (a); weakly identified. Within-cohort changes mix career and calendar time. 95% intervals, types stated in each panel.

Table 3: Career-time summary. US: PSEO release V4.14.1, on which every row is available (the canonical V4.13.0 gives slopes within 0.004 per year of these; SI Section S6); bachelor’s graduates, field prestige F , per-field slopes averaged over fields, two-way (field, institution) cluster intervals unless marked; balanced fixed cohorts 2001–03 to 2010–12 unless the row states another sample. UK: LEO, balanced fixed cohorts 2013/14–2016/17, academia-wide prestige G ; two-stage bootstrap over subjects and providers. Slopes are changes in coupling (or in a loading) per year since graduation. 95% intervals.

Quantity	US, V4.14.1 (40 fields) ^a	UK (33 subjects)
Coupling at first / middle / last horizon	+0.159 / +0.294 / +0.433	+0.379 / +0.438 / +0.484
Within-cohort slope (crossed bootstrap)	+0.030 [+0.020, +0.041] ([+0.018, +0.043])	+0.026 [+0.014, +0.037]
Calendar-matched contrast; US without earnings years 2008–10, 2020–22 (38 fields)	+0.015 [+0.007, +0.023]; +0.014 [+0.005, +0.024]	+0.031 [+0.021, +0.042]
Cross-cohort drift at entry	+0.018 [+0.010, +0.025]	−0.006 [−0.015, +0.003]
Career-time bound if calendar, cohort effects ≥ 0 ^b	[+0.015, +0.032]	—
17-cell surface (24 fields): θ_h / θ_c	+0.031 [+0.019, +0.043] / +0.015 [+0.008, +0.023]	—
Coverage control: share of slope removed (US); slope net of coverage (UK)	0.195 [0.097, 0.293]	+0.017 [+0.005, +0.027]
Slope net of coverage and research-doctorate rate (144 cells; raw +0.0305)	+0.0103 [+0.0027, +0.0178]	—
Slope net of entry-year SAT and admission rate (39 fields, 109 cells; raw +0.0289)	+0.0112 [+0.0026, +0.0198]	—
Prior-attainment standardised slope; with G corrected for validity 0.74	—	+0.028 [+0.016, +0.038]; +0.035 [+0.019, +0.051] ^c
Slope of b_G (d_G) / of b_F (d_F)	+0.035 [+0.019, +0.052] / +0.000 [−0.013, +0.014]	—
$d_G - d_F$ (crossed bootstrap)	+0.035 [+0.007, +0.063] ([−0.003, +0.070])	—
Loading on G net of selectivity (2026 controls): rank regression / partial correlation (37 fields)	+0.016 [−0.011, +0.044] / +0.014 [+0.003, +0.024]	+0.008 [−0.006, +0.022] ^c

^a The calendar-matched contrast, the drift and the bound use the 39 fields with triple-matched institutions; rows naming another sample use it. ^b Both ends would be biased upward if graduate study depressed the early medians of higher-status institutions more than others’. ^c Two-way (subject, provider) interval.

with these contrasts; whether licensure itself raises wages is contested (Kleiner and Krueger, 2013; Redbird, 2017), and these data isolate neither reading (SI Section S7).

Quantities. Where pay follows a schedule the wage stops revealing the market’s valuation (Proposition 5), so weak coupling there says little about whether the market’s ordering agrees with the academy’s; the model’s content is about quantities. In education and in health professions (CIP-2 51; PSEO Flows do not release nursing separately), scored on teachers’ and registered nurses’ pay, G orders institutions by the scheduled pay level of the census divisions where their graduates work five years after graduation (mean Spearman +0.238 [+0.065, +0.380]; T6, supported), mainly in health professions (+0.319 [+0.159, +0.451], 116 institutions; education +0.157 [−0.058, +0.330]). The index uses nominal division wages and largely records the institution’s own division, so it does not separate status from location (SI Section S11).

6 Discussion

Where the two orderings agree. The academy’s hiring order and the market’s pay order of the same institutions agree in most fields, increasingly so with years since graduation. Where they agree, they agree at the institution’s name, mostly in what both share with the selectivity of its

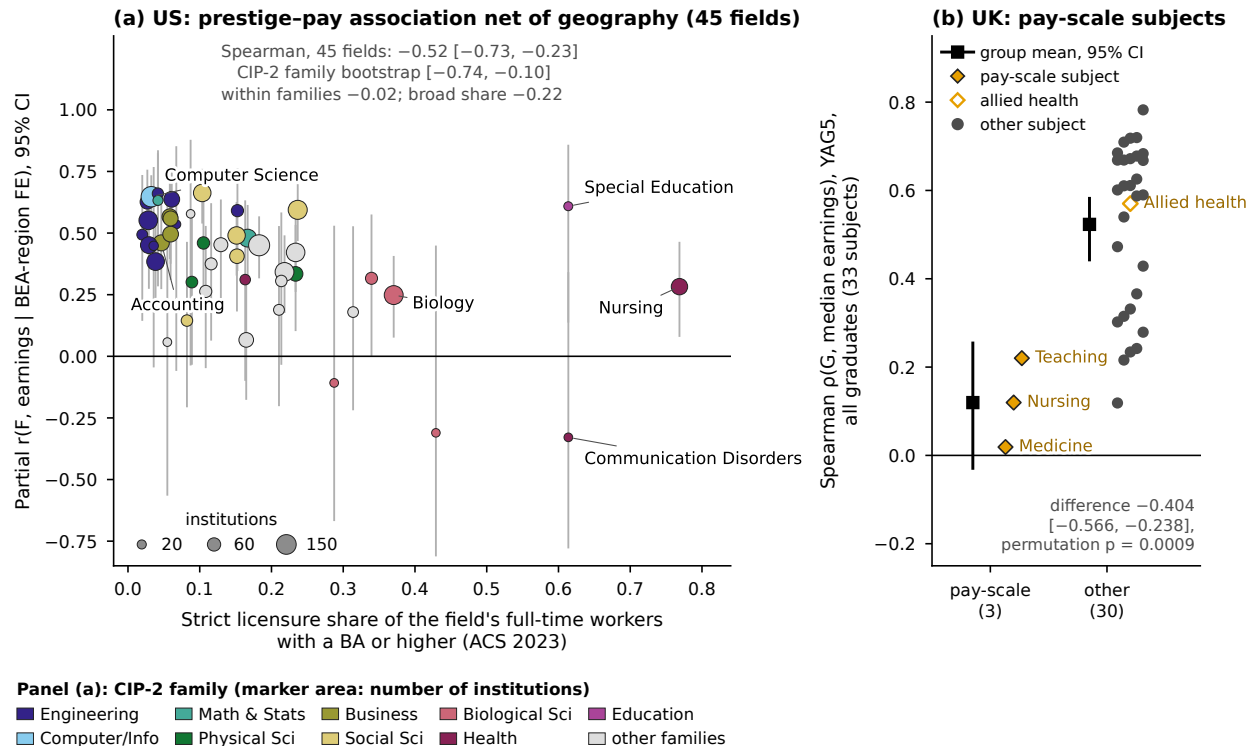


Figure 5: Where coupling is weak. (a) US: partial correlation of F with Scorecard four-year earnings net of BEA-region effects (one of six sample-by-geography versions) against the field's share of full-time bachelor's holders in strictly licensed occupations (ACS 2023); one point per field, coloured by CIP-2 family, sized by institutions. (b) UK: coupling of G with LEO five-year earnings (33 subjects) in the three pay-scale subjects, allied health and the others; group means. 95% intervals.

intake; below the name they agree little, clearly only in computer science, with about 0.8% of pay per SD of department prestige (up to about 3% under the lower-bound reliability of F). The pay in these statements is not netted of destination location beyond a check on five-year coupling (T7), and about a fifth of pay coupling runs between employer sectors (T8), which counts as the market's valuation only if employers ration access to sectors. Because G pools the same hiring data over all fields, "the name" here is the part of the hierarchy shared across an institution's departments.

For the science of science. The faculty-hiring hierarchy is a reliable measure of position inside the academy, the academia-wide ordering especially, but it is a net-placement position that partly reflects doctoral production (the audit bounds that share from above), not one index of a department's standing as producer and as employer of PhDs. Studies that use hiring-network prestige as a proxy for a program's training quality or labour-market standing should expect it to behave mostly as a measure of institutional status. The one field-specific exception, computer science, fits selection into the major as well as department standing.

For the college-quality literature. A hiring-network axis shows the pattern Britton et al. (2022) report: most of the prestige ordering of pay goes with selectivity (US $+0.43$ to $+0.14$ across about 50 fields; UK $+0.50$ to $+0.13$ on 32 subjects), and what remains beyond selectivity and academia-wide status is small. Unlike their league-table result, 11 of 46 US fields keep a partial

coupling more than 1.96 standard errors above zero net of the broad selectivity set (without G). The cross-field ordering of coupling is nearly collinear with how steeply each field’s pay follows institutional selectivity (Bloem et al., 2024), as one dominant institution factor would produce, so at the resolution of public data “the status gradient in pay differs by field” and “the department-prestige gradient in pay differs by field” cannot be told apart. In the UK the within-band premium to intake selectivity does not decay on average over bands within fixed cohorts (it rises), so we do not describe it as reputation that learning corrects; the top band, where own attainment is most uniform, cannot answer.

Career time. Within fixed cohorts coupling rises by about +0.03 per year in both countries. In the US its career-time part is bounded, [+0.015, +0.031] per year, only under a sign assumption; about three fifths of the within-cohort rise is shared with selectivity, and a fifth to a third moves with where graduates work. Net of selectivity the status loading does not decline in either country, but under rational expectations about names that employers see often that is close to guaranteed, so it is not evidence about how employers learn about individuals. Set against small or fading causal selectivity premia (Dale and Krueger, 2014; Mountjoy and Hickman, 2021), the most parsimonious reading of the rise is residual selection; it also fits status-correlated growth, sorting into steeper careers and the timing of graduate study, and it is not detectably concentrated where employers rarely see an institution’s graduates. The data do not distinguish these readings.

Below the name. Why agreement below the name is small is not identified. Department standing may carry little information about graduates’ market productivity, or employers may not attribute performance to departments and learn slowly about individuals. If learning speeds estimated in other populations (Lange, 2007; Arcidiacono et al., 2010) applied here, the small four-year remainder would imply little departmental content; nothing in these data tests this, and no test here separates the two readings.

What would decide the open questions. Four kinds of data would separate department standing from program-level selection: program-level entry attainment, which no public US file has; department ratings from different assessment rounds (National Research Council, 1995, 2011) paired with PSEO cohorts; admission routes into majors for a larger sample; and linked student records with entry scores, major and earnings, the decisive design, which are not public. Census-level hiring edges would narrow the reliability bracket of F and allow a degree-corrected ranking without the single-hire suppression. Two re-estimates are feasible with the public files already used: the within-institution and computer-science analyses with a join by institution identifier and entry-year SAT averages, and the status-loading test with entry-year selectivity. Graduate enrolment by institution, field and cohort would test the graduate-study reading of the career rise.

7 Limitations

Each of the following limits a specific claim. *Theory.* The model was written after most results had been seen; it organises them, and they are not evidence for it. Its nine tests were specified before the analysis but not publicly registered, and they use data that overlap those of results already seen (the pay side of T2 was known, T3 was chosen knowing that the G loading rises, T4 knowing the within-band rise of raw coupling, T5 had a known selectivity analogue, and T6–T8 reuse PSEO Flows data analysed in script 61); T2 and T5 are inconclusive for lack of power, and the amended T1 verdict rests on a calibration chosen after the literal one had degenerated (SI

Section S11). *Medians and coverage.* Scorecard medians describe Title IV-aided completers who work and are not enrolled; PSEO lacks most of the top prestige quartile, and neither source sees the upper tail within a program. *Sample.* The within-institution and computer-science samples lack 18 to 21 of the 50 highest-ranked computer-science departments (Section 5.2). *Selectivity proxies.* Selectivity is institution-level; re-dating it to entry years moves none of the three primary statistics detectably, but cannot show that the proxies capture student quality, and whatever they miss stays in every partial coupling; the status loadings, the +0.059 remainder and the +0.94 use 2026 controls. *Location.* Pay is not netted of destination location: only raw five-year coupling on the PSEO panel was checked net of the wage level of graduates' destination census divisions (unchanged, T7); the Scorecard estimates (the control ladder, the partial couplings and the within-institution slope) were not re-estimated net of destination location, location within divisions is not netted, and the local-demand control describes the institution's state. *Earnings reliability.* The noise propagation is secondary, not yet independently verified, and its error model is approximate. *Timing.* Career dynamics are not separable from calendar time without a sign assumption, or from the timing and returns of graduate degrees, which could bias both ends of the bound upward. *Prestige.* The reliability of the published field ranks can only be bracketed (0.81–0.96), so the size of the remainder is uncertain (+0.0098 to +0.0297 disattenuated); F is partly production volume, a share the audit bounds only from above, since neither of its branches isolates production; and the UK hierarchy rests on self-reported ORCID careers of uncertain validity. *Rank statistics.* Low coupling in UK pay-scale subjects may reflect little cross-provider pay variation. *Descriptive design.* Nothing here separates value added, selection and employer screening, and beliefs about prestige are not measured. *Field-level inference.* Moderators measured once per field rest on 40–50 fields and about 18 CIP-2 families. *Power.* Nulls are statements of what the data cannot detect; SI Section S9 gives the power or interval of the main nulls. *Specification.* The canonical specification was fixed after the main analyses had been run; SI Section S4 shows how the main statistics move with the alternatives.

8 Methods

Within-institution model. For the program of field f at institution i ,

$$\log Y_{if} = \alpha_i + \gamma_f + \beta z(F)_{if} + \sum_{f'} D_{f'}(d_{f'} z\text{SAT}_i + a_{f'} z\text{ADM}_i + p_{f'} z\text{Pell}_i + t_{f'} zG_i) + e_{if}, \quad (3)$$

where $z(\cdot)$ standardises within field, α_i and γ_f are institution and field fixed effects, and $D_{f'}$ indicates field f' . The field-specific slopes let the earnings gradients in selectivity, Pell share and G differ by field, so β is identified only from departments whose standing differs from what their institution's traits imply. Institutions with one field and fields with fewer than 15 programs are dropped iteratively; intervals come from an institution cluster bootstrap with 999 draws shared across specifications, so that differences are paired (SI Section S4).

Career-time contrasts. On institutions observed in all three cells, the within-cohort change $(y_{10} - y_1)/9$, the calendar-matched contrast (cohort c at y_{10} against cohort $c+9$ at y_1 , over 9) and the cross-cohort drift at y_1 equal $a+b$, $a-g$ and $b+g$ in a linear model with career, calendar and cohort effects a , b and g ; on the 17-cell surface, the within-window and across-window slopes equal $\theta_h = a+b$ and $\theta_c = b+g$. Only under $b, g \geq 0$ is the career-time effect bounded: $a \in [a-g, a+b]$, that is, between the calendar-matched contrast and the within-cohort change, or $a \in [\theta_h - \theta_c, \theta_h]$ on the surface (Fosse and Winship, 2019b). No point estimate of a is reported (SI Section S6).

Further details. The SI documents data (S1), prestige measures and the audits of F (S2), the coupling estimator (S3), selectivity designs (S4), the UK analysis (S5), career time (S6), licensing and employer sector (S7), tail outcomes (S8), power and the hypotheses examined (S9), earnings as a placement measure (S10), and the model, its mapping of results and tests T0–T8 (S11).

Declarations

Availability of data and materials. All inputs are public: the faculty-hiring data (Wapman et al., 2022a); College Scorecard field-of-study, institution and historical institution files (U.S. Department of Education, 2026); Census PSEO earnings and flows, releases V4.13.0 (2025Q4) and V4.14.1 (2026Q2) (U.S. Census Bureau, Longitudinal Employer-Household Dynamics, 2025, 2026); the Department for Education’s “Graduate outcomes (LEO): provider level data”, 2022–23 release (Department for Education, 2026); the ORCID-derived academic mobility edges (version 20260419, doi:10.5281/zenodo.19651302; Li, 2026), built from ORCID public data (ORCID, Inc., 2026), with institutions resolved through the ROR data dump (v2.8, doi:10.5281/zenodo.20512981; Research Organization Registry, 2026); Opportunity Insights college tables, including the Barron’s 2009 selectivity index (Chetty et al., 2020); the 2023 American Community Survey public-use microdata (U.S. Census Bureau, 2026); the Survey of Earned Doctorates by baccalaureate institution (National Center for Science and Engineering Statistics, 2026) and IPEDS completions and directory files (National Center for Education Statistics, 2026); state unemployment rates from the Bureau of Labor Statistics Local Area Unemployment Statistics (U.S. Bureau of Labor Statistics, 2026); occupational prestige ratings (Hughes et al., 2024; Condon and Hughes, 2022) and the New York Fed college labour-market data by major (Federal Reserve Bank of New York, 2026); and official university admissions pages, saved with their access dates. Provenance for every input (URL, version, access date and licence) is recorded in `data/raw/SOURCES.md` of the repository. No input dataset is redistributed; the derived tables are regenerated by the scripts. The hand-coded computer-science admission file (URL, access date and verbatim sentence for each program), which no script can regenerate because the pages change, and the small derived tables (UK prestige ranks, CIP map) are available from the author.

Code availability. The numbers in this article come from seeded scripts (scripts 01–75; mainly 52–75). The figures, and the numbers marked “SI check”, come from `scripts/70_paper_figures.py` (with `-si-checks`); the SI names the source of every other number. The repository is at <https://github.com/soray42/degrees-of-separation>. At the time of writing, scripts 67–69 and 72–75 and the current versions of scripts 65 and 70 are not yet in the public repository; they will be added, with the digests of their pre-specification blocks, in a tagged release. The binding texts of tests T0–T8 are reproduced in the docstring of script 75 with their SHA-256 digests; no public time-stamped deposit was made before the tests were run.

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